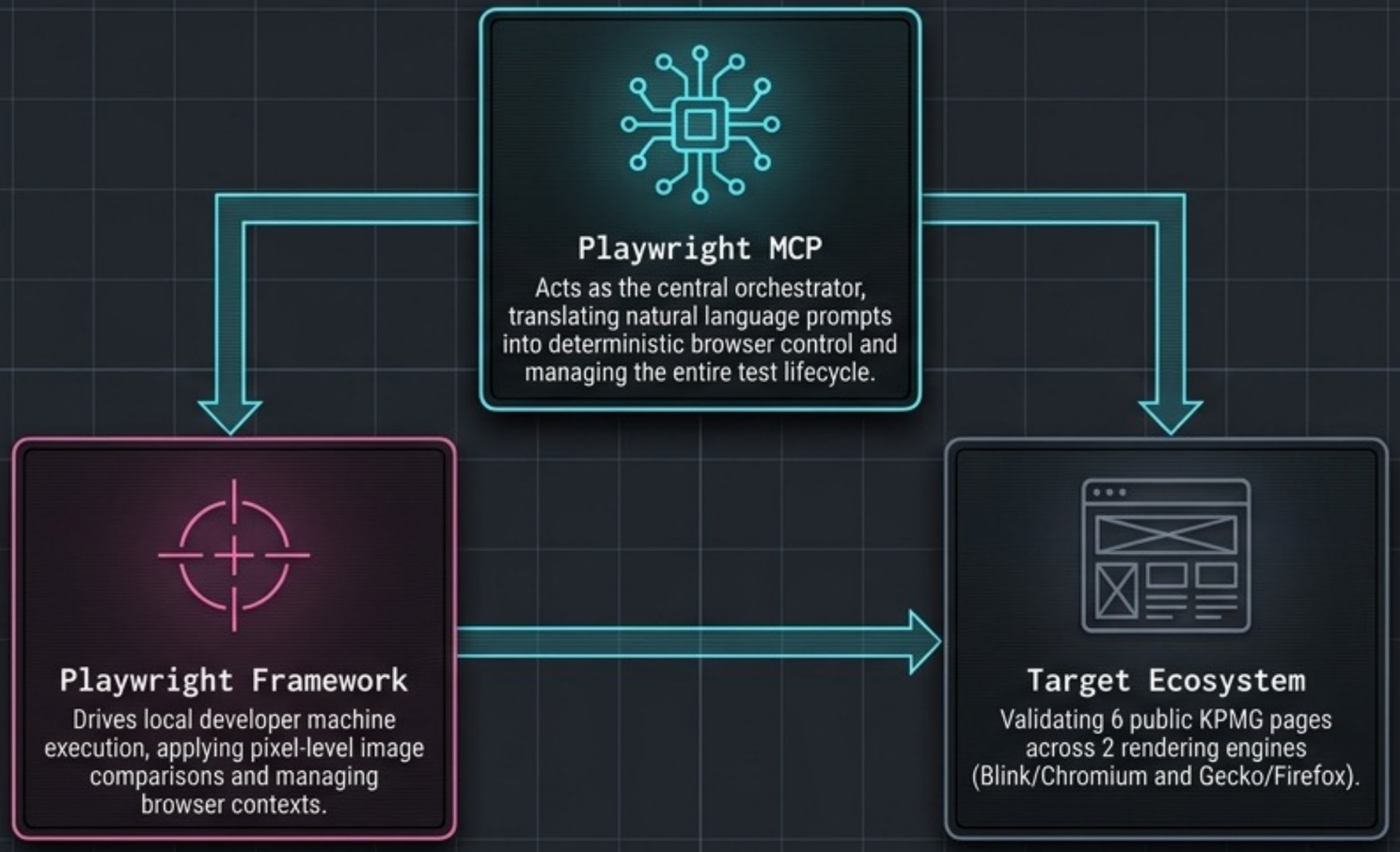
Visual Regression Testing Agent Architecture

An AI-powered framework using Playwright and Model Context Protocol (MCP) to automate deterministic visual testing for KPMG web pages.



[SCOPE] 6 Pages × 2 Browsers × 1 Viewport (1920×1080) = 12 Total Baseline Screenshots

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Autonomous Orchestration Pipeline

1 Navigation

MCP-Driven Orchestration

MCP reads URL from JSON configuration and orchestrates initial routing to target KPMG environments without requiring authentication headers.

2 Page Stabilization

Cookie Dismissal & CSS Animation Disabling

Wait for networkidle. Programmatically reject GDPR consent banners, disable all CSS transitions, and trigger lazy-loaded elements via full-page scroll.

3 Screenshot Capture

Full-Page Stitching & Masking

Capture complete 1920x1080 document height. Apply configured maskSelectors to block out dynamic content (like changing timestamps) to prevent false positives.

4 Pixel-Level Comparison

Baseline Validation

Evaluate actual capture against Git-stored baselines using a highly precise 0.5% maximum diff tolerance.

5 Reporting

Visual Diff Generation

Output an actionable HTML report featuring a custom visual diff panel alongside JSON-based historical trend tracking.

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Balancing Execution Speed with Architectural Resilience

Machine Capabilities (Autonomous Resilience)

Automation of Manual Inspection

Replaces error-prone, time-consuming manual QA with consistent, deterministic visual validation across multiple rendering engines (Chromium and Firefox).

Progressive Stabilization

Utilizes a smart 3-attempt retry matrix. If standard stabilization fails, the system escalates wait times, forces CSS injections, and applies hard settle delays to defeat flaky page loads.

Human Safeguards (Strict Authority)

Interactive Human-in-the-Loop

Baseline updates are never completely autonomous. The custom diff-reviewer utility mandates explicit, interactive developer approval to prevent accidental overwrites of the golden baselines.

Precision Visual Debugging

Failures generate side-by-side comparison panels (Baseline vs. Actual vs. Diff Overlay), illuminating the exact disparate pixels so engineers can diagnose regressions instantly.

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[VALIDATION] Consecutive runs with no code changes consistently produce zero diffs.
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